PV Circonomy

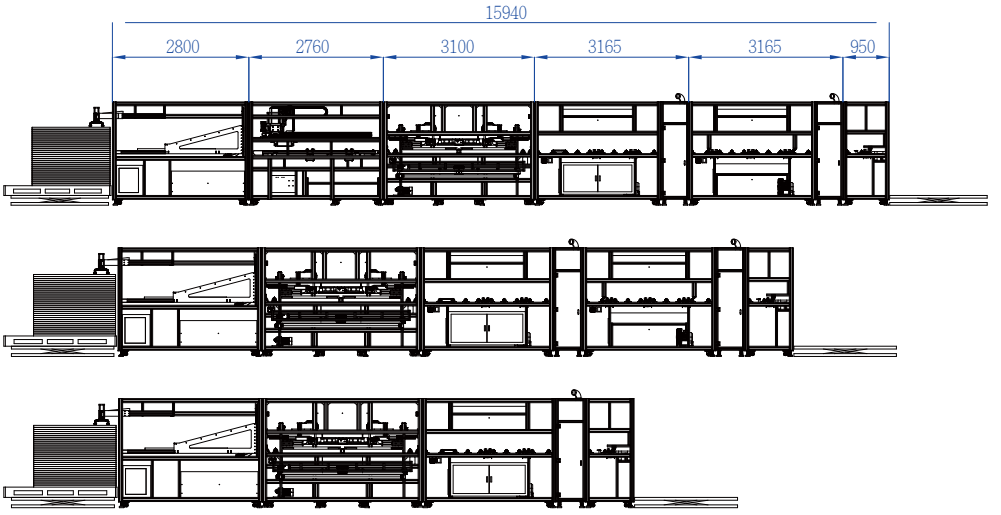
PEOPLE
TECHNOLOGY
MATERIALS
IN CIRCULATION

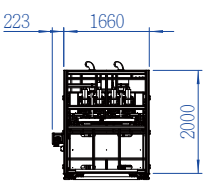
PV Circulator®

PV RECYCLING SOLUTION

One platform
Any feedstock
Full material value.

One integrated system for intact, broken and bifacial EoL PV modules.





MODULAR | DEPLOYABLE | HIGH-VALUE | SUSTAINABLE

PV CIRCULATOR SOLUTION

One system. Any feedstock.

Process intact, broken and bifacial PV modules with one integrated recycling platform.



INTACT

Standard EoL modules from solar farms, rooftops and repowering projects.

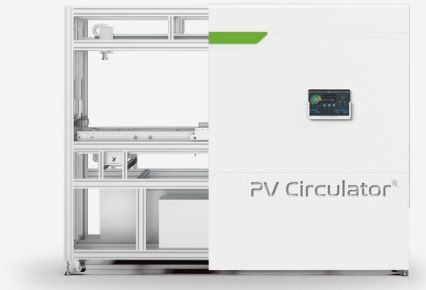


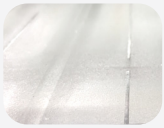
BROKEN & BIFACIAL

Broken, shattered and glass-glass bifacial modules.

ONE PV CIRCULATOR PLATFORM

Configured around your feedstock and operation.





GLASS



ALUMINUM



JB & CABLE



POLYMER

EVA/Backsheet

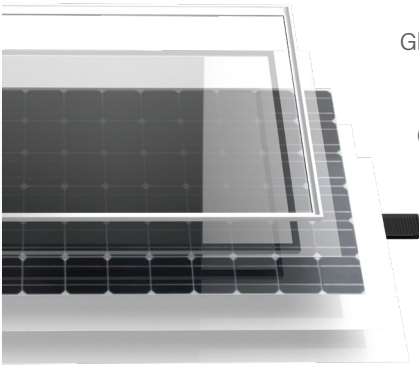


CELLs

FROM RECYCLING TO REAL VALUE

More than a processing fee.

Layer-by-layer separation minimizes cross-contamination, preserving material quality and value.

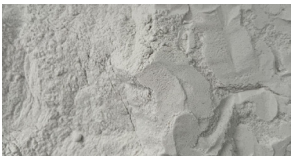


	Conventional	PV Circulator
Glass	contaminated	intact & clean
Cell material	contaminated	concentrated
EVA	eliminated	recovered
Backsheet	eliminated	recovered

PRESERVE MATERIAL VALUE

Cleaner separation. Greater value potential.

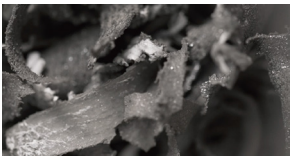
GLASS



Clean, not mixed

More options for reuse, recycle and pozzolan applications.

POLYMERs



Recovered, not destroyed.

More options for offtake and circular use.

CELLs



Concentrated, not dispersed.

More potential Ag/Si recovery and greater value.

GO FURTHER WITH DOWNSTREAM SOLUTIONS.

PV Circonomy extends value through purification and offtake.



Cell Purification

Equipment & Processing Services

Recover Ag and Si from separated cell material.

47	14
Ag	Si
Silver	Silicon
107.8682	28.086



Polymer Offtake

A circular destination for recovered PV polymers.

- ✓ No landfill required
- ✓ No additional treatment fee
- ✓ Support circular use

BUILD FOR REAL-WORLD OPERATIONS

Engineered for scale.

ECO
MODULAR
AUTOMATED
DEPLOYABLE

A modular, automated, dry physical PV recycling system designed for continous, efficient and reliable operation.



1	>99	<1	1	$0.05
panel/ min throughput	% material recovery	kWh per panel	Person operation	per panel consumables cost

A CONTINOUS, LAYER-BY-LAYER PROCESS

1. **Infeed**
   Automated continuous loading, 30+ panels per pallet
2. **Identification & Measurement**
   Edge AI identification, automatic parameter setup and web3-enabled module-level digital records
3. **Junction Box Removal**
   Automatic removal of single or multiple junction boxes
4. **Frame Removal**
   Adaptive dismantling to different module sizes
5. **Delamination 1**
   Room-temperature backsheet separation
6. **Delamination 2**
   Room-temperature, layer-by-layer cells and encapsulant separation with minimal cross-contamination and preserving material value
7. **Outfeed**
   Brush-wheel cleaning removes surface residue for cleaner recovered glass
8. **Discharge**
   Automated continuous glass discharge

BUILT FOR MORE THAN JUST RECYCLING

Edge AI Automation

Automatic identification and process control

Digital Traceability

Module-level processing record and LCI data

Deployable Design

Modular system with setup and teardown within 24 hours

Low Environment Impact

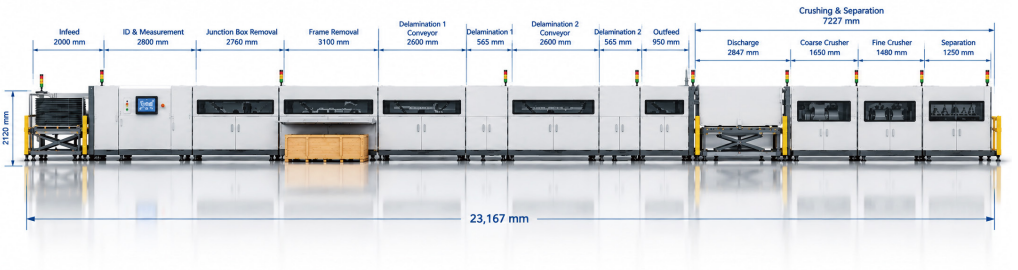
No water
No chemicals
No thermal treatment

CONFIGURED FOR YOUR OPERATION

Choose the right fit.

Three automation levels. Two feedstock capabilities. One platform.

	Economic essential processing	Basic complete separation	Premium automated & traceable
InFeed	○	●	●
ID & Measurement			●
JB Removal			●
Frame Removal	○	●	●
Delamination 1	○	●	●
Delamination 2		●	●
OutFeed	○	●	●
Discharge	○	●	●
Throughput	1 panel/min	1 panel/min	1 panel/min
Traceability			●



PLUS | EXPANDED FEEDSTOCK CAPABILITY

Process broken & bifacial modules across all three automation levels.

Coarse Crusher	○	●	●
Fine Crusher	○	●	●
Separation	○	●	●
Dust Control	Enclosed with dust extraction and dedicated collection		

ECONOMIC PLUS | BASIC PLUS | PREMIUM PLUS

Built to perform. Support to last.

TECHNICAL BENCHMARK

A cleaner, smarter approach to PV recycling.

	PV Circulator	Typical Competitors
Separation	Layer-by-layer	Crushing & sorting
Labor requirement	1 operator	Multiple operators
Consumables cost	As low as $0.05/panel	Process dependent
Material recovery rate	>99%	<90%
Cross-contamination	Very low	High
Glass quality	High	Lower
Cell material	Concentrated	Dispersed by glass
Polymer recovery	Yes	No
Secondary waste	Minimal	Dust / mixture
Energy use	< 1kWh/panel	Process dependent
Deployment	Modular / deployable	Typically fixed
Material value	High	Limited by mixture

LIFECYCLE SUPPORT

From installation to optimization, supports your operation beyond delivery.

Installation & Commissioning

Professional installation and system commissioning for stable startup.

Operator Training

Hands-on training and SOP guidance for safe, efficient operation.

Remote Support

Remote diagnostics and troubleshooting to minimize downtime.

Spare Parts & Maintenance

Parts supply and preventive maintenance support.

Process Optimization

Ongoing process and software optimization as operation evolve.

Downstream Support

Material purification and offtake solution beyond recycling.

CLEAN ENERGY
DESERVES A CLEANER ENDING.

Circular today Brighter tomorrow

PV Circonomy is committed to creating a closed-loop solar industry through innovative recycling solutions, material upcycling and strategic partnerships.

CIRCULAR MATERIALS REIMAGINED

LOWER CARBON FOOTPRINT

A MORE SUSTAINABLE FUTURE



Let's build a more circular solar future

Partner with PV Circonomy
to turn PV waste into valuable resources.



Consultation
Discuss your feedstock, capacity and project goals.



System Configuration
Tailored solutions for your operation.



Site Planning
Layout, utilities and deployment options.



Lifecycle Support
Installation, training, spare parts and remote support.



Material Offtake
Access downstream material partners, including PV Circonomy polymer offtake.

Contact Us

Connect with PV Circonomy

Explore your PV recycling business case, request a quotation or schedule a meeting.

OPEN OUR ONLINE CONTACT FORM

PV Circonomy | California, USA

PV Circonomy

CIRCULAR FOR A BRIGHTER TOMORROW